Project 2: SOC- EDR Grid

1. Project Overview

This project builds a **Security Operations Center (SOC) monitoring environment** using **Wazuh**. The objective is to detect, monitor, and analyze security events from endpoints in real time.

The system follows a **centralized EDR + SIEM architecture** where all endpoint machines send their logs to a central server for analysis.

The log flow is:

Kali Linux (Attacker)

↓

Ubuntu Desktop (Victim)

Wazuh Agent

↓

Ubuntu Server

Wazuh Manager + Filebeat + Indexer

↓

Wazuh Dashboard

2. SOC (Security Operations Center)

A **Security Operations Center (SOC)** is a centralized team or system responsible for monitoring and defending an organization's IT infrastructure.

Main responsibilities of SOC include:

• Continuous monitoring of systems (24×7)
• Detecting cyber attacks in real time
• Investigating suspicious alerts
• Responding to security incidents
• Maintaining system security posture

The SOC uses tools like **Wazuh SIEM** to analyze security logs and identify threats.

3. EDR (Endpoint Detection and Response)

EDR (Endpoint Detection and Response) focuses on monitoring endpoint devices such as:

• Servers
• Workstations
• Virtual Machines
• Cloud instances

The purpose of EDR is to detect suspicious behavior occurring directly on the endpoint.

In this project, **Wazuh Agent acts as the EDR component**.

The agent collects:

• System logs
• Authentication logs
• File integrity changes
• Malware activity
• Unauthorized access attempts

These logs are then sent to the Wazuh Manager for analysis

4. Ubuntu Server – Wazuh Manager (SOC Server)

The Ubuntu Server acts as the central brain of the SOC environment.

Components installed on the server include:

• Wazuh Manager: Without the Wazuh Manager, agents cannot communicate or send logs.
• Wazuh Indexer: The Wazuh Indexer is responsible for storing security logs.
• Filebeat: Filebeat is a log forwarding tool. Enable data visualization on the dashboard
• Wazuh Dashboard

Conclusion:

• Endpoint monitoring using Wazuh Agent
• Centralized log analysis using Wazuh Manager
• Security event visualization through the Wazuh Dashboard
• Real-time detection of potential threats

Figure 1: Installation process

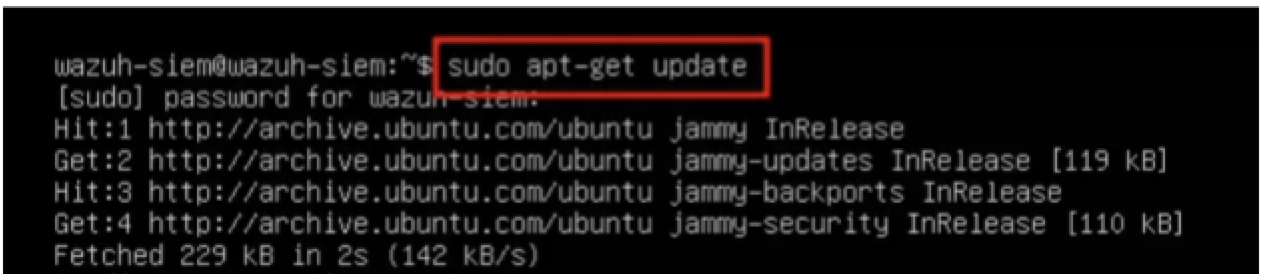
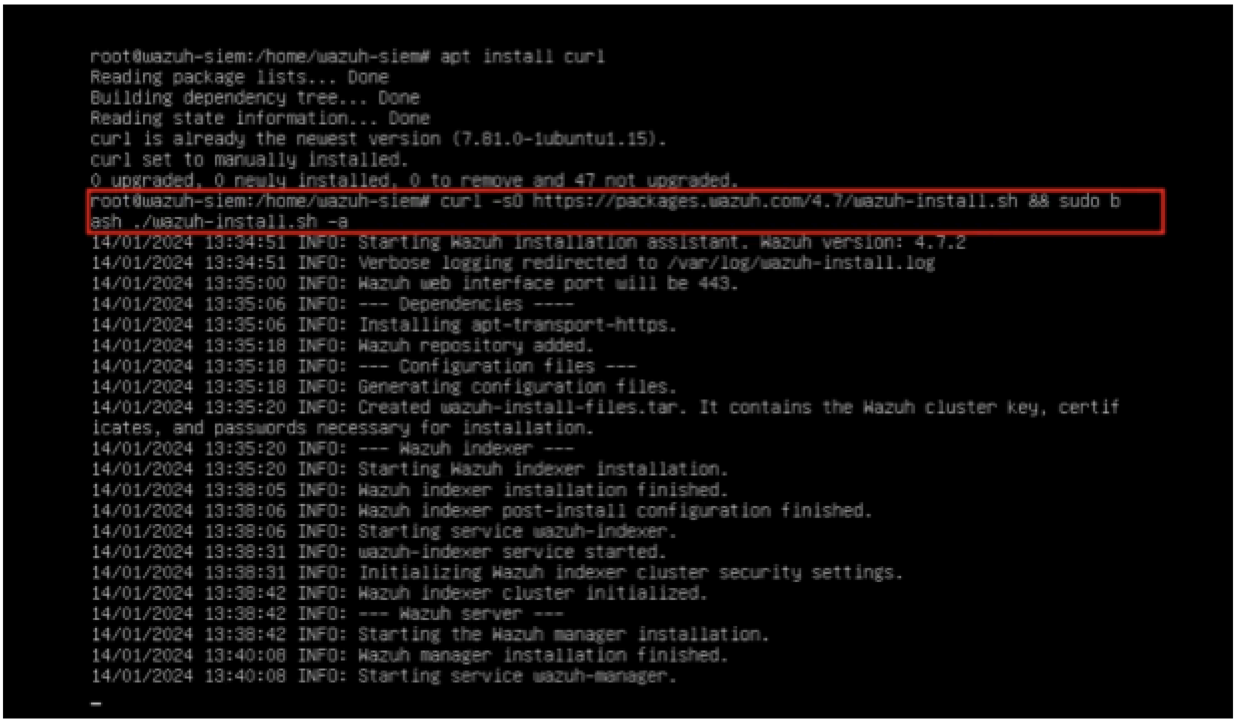
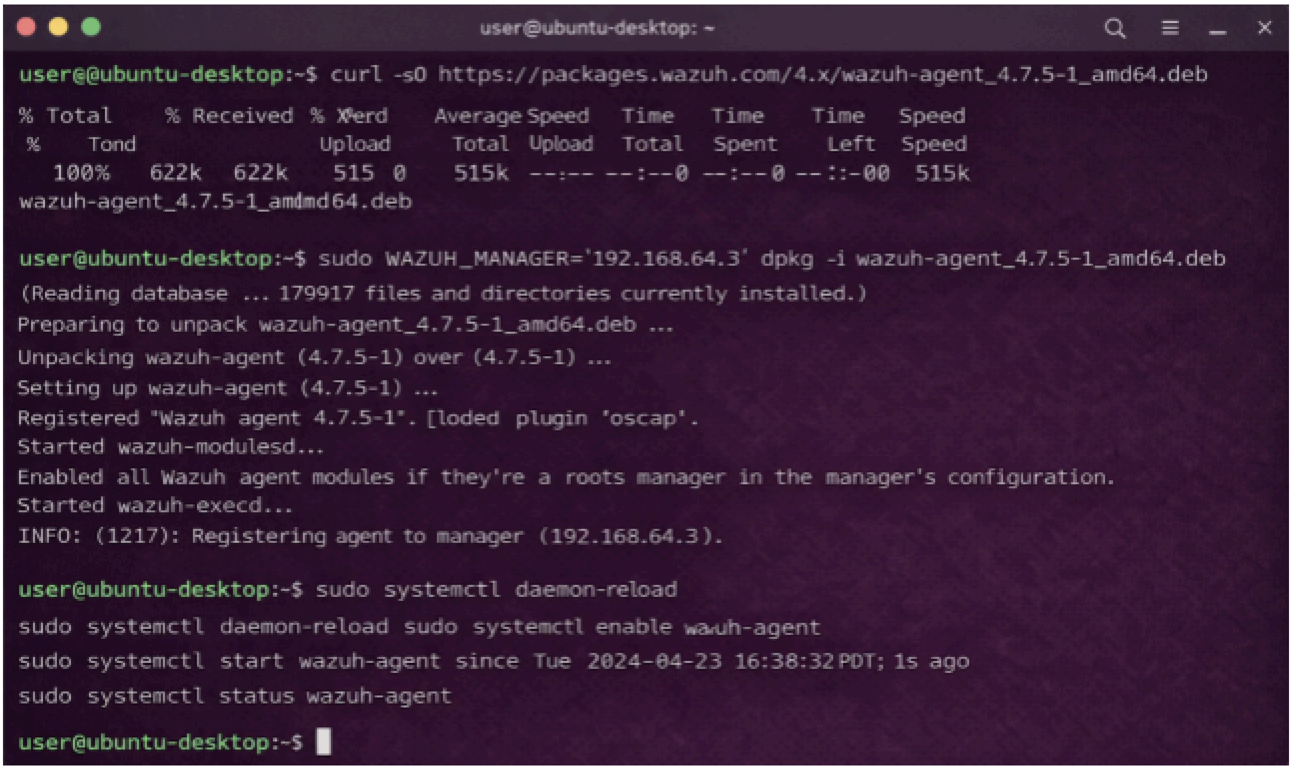


Figure 2: setup of Wazuh



- Used to **install and connect the endpoint (Ubuntu Desktop) to the Wazuh Manager server.**

Figure 3: configure the Wazuh Agent on Ubuntu Desktop



- The Wazuh agent is installed on the Ubuntu Desktop endpoint to collect security logs and send them to the Wazuh Manager. This enables centralized monitoring, threat detection, and incident response in the SOC environment.

Figure 4: Certificates creation

```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# curl -sO https://packages.wazuh.com/4.7/wazuh-certs-tool.sh
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# curl -sO https://packages.wazuh.com/4.7/config.yml
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# ls
config.yml  wazuh-certs-tool.sh
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#
```

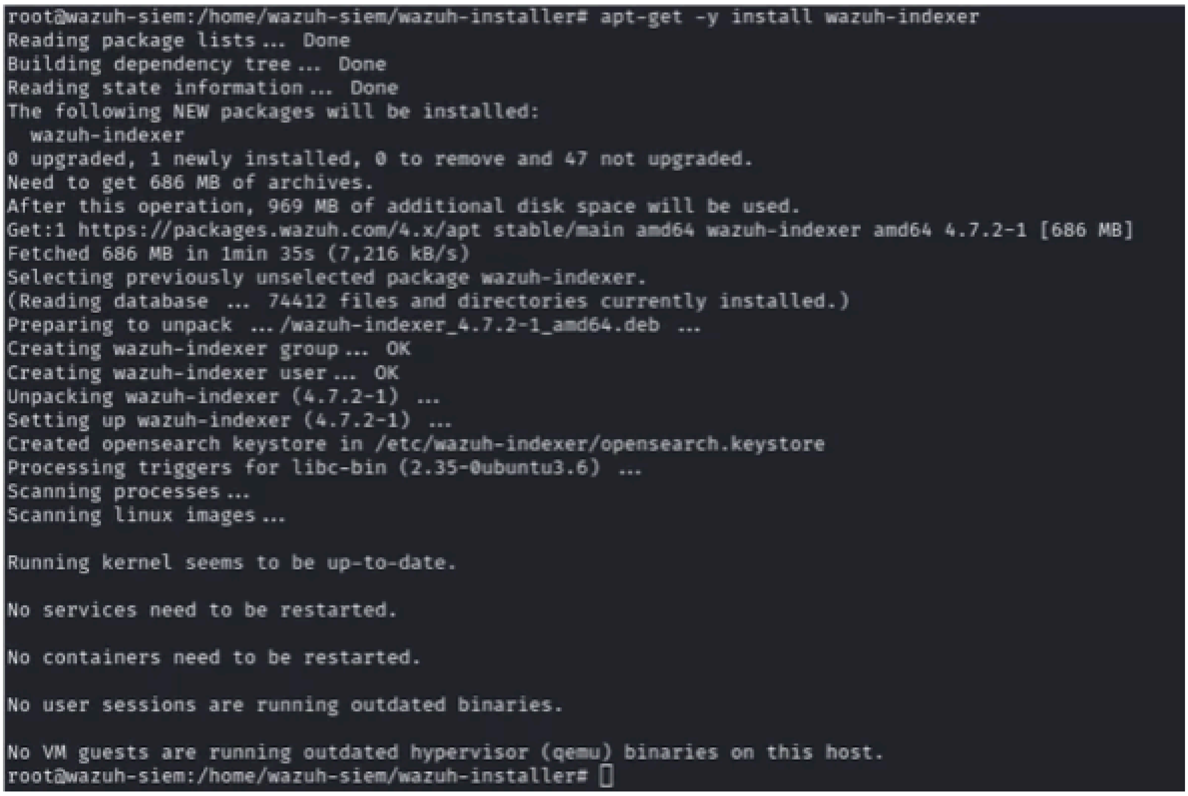
- Certificates are using for **encryption** the content.

Figure 5: ip a command to retrieve your server's IP

```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:8c:1d:0b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.251.150/24 metric 100 brd 192.168.251.255 scope global dynamic ens33
       valid_lft 1016sec preferred_lft 1016sec
    inet6 fe80::20c:29ff:fe8c:1d0b/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:f8:c9:34:37 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
6: vxlan.calico: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UNKNOWN group default
    link/ether 66:06:41:a0:3b:ce brd ff:ff:ff:ff:ff:ff
    inet 10.1.26.64/32 scope global vxlan.calico
       valid_lft forever preferred_lft forever
    inet6 fe80::6406:41ff:fea0:3bce/64 scope link
       valid_lft forever preferred_lft forever
7: calid6fabdc3875@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UP group default
    link/ether ee:ee:ee:ee:ee:ee brd ff:ff:ff:ff:ff:ff link-netns cni-d61607d5-aa2a-161d-86bb-50a27eaed12b
    inet6 fe80::ecee:eeff:feee:eeee/64 scope link
       valid_lft forever preferred_lft forever
8: calic54cd419f2e@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UP group default
    link/ether ee:ee:ee:ee:ee:ee brd ff:ff:ff:ff:ff:ff link-netns cni-5e65c97a-698f-41ce-6820-75cf85c03854
    inet6 fe80::ecee:eeff:feee:eeee/64 scope link
       valid_lft forever preferred_lft forever
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#
```

```
 GNU nano 6.2
nodes:
  # Wazuh indexer nodes
  indexer:
    - name: node-1
      ip: "192.168.251.150"
    #- name: node-2
    #  ip: "<indexer-node-ip>"
    #- name: node-3
    #  ip: "<indexer-node-ip>"

  # Wazuh server nodes
  # If there is more than one Wazuh server
  # node, each one must have a node_type
  server:
    - name: wazuh-1
      ip: "192.168.251.150"
    #  node_type: master
    #- name: wazuh-2
    #  ip: "<wazuh-manager-ip>"
    #  node_type: worker
    #- name: wazuh-3
    #  ip: "<wazuh-manager-ip>"
    #  node_type: worker

  # Wazuh dashboard nodes
  dashboard:
    - name: dashboard
      ip: "192.168.251.150"
```

Figure 6:Installing Wazuh indexer.

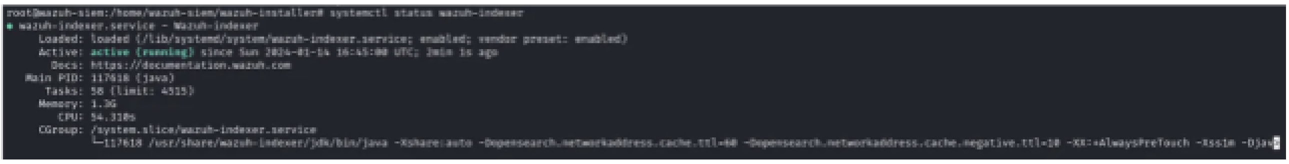


```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# apt-get -y install wazuh-indexer
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wazuh-indexer
0 upgraded, 1 newly installed, 0 to remove and 47 not upgraded.
Need to get 686 MB of archives.
After this operation, 969 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-indexer amd64 4.7.2-1 [686 MB]
Fetched 686 MB in 1min 35s (7,216 kB/s)
Selecting previously unselected package wazuh-indexer.
(Reading database ... 74412 files and directories currently installed.)
Preparing to unpack .../wazuh-indexer_4.7.2-1_amd64.deb ...
Creating wazuh-indexer group... OK
Creating wazuh-indexer user... OK
Unpacking wazuh-indexer (4.7.2-1) ...
Setting up wazuh-indexer (4.7.2-1) ...
Created opensearch keystore in /etc/wazuh-indexer/opensearch.keystore
Processing triggers for libc-bin (2.35-0ubuntu3.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# 
```

Figure 7:Configuring Wazuh indexer.

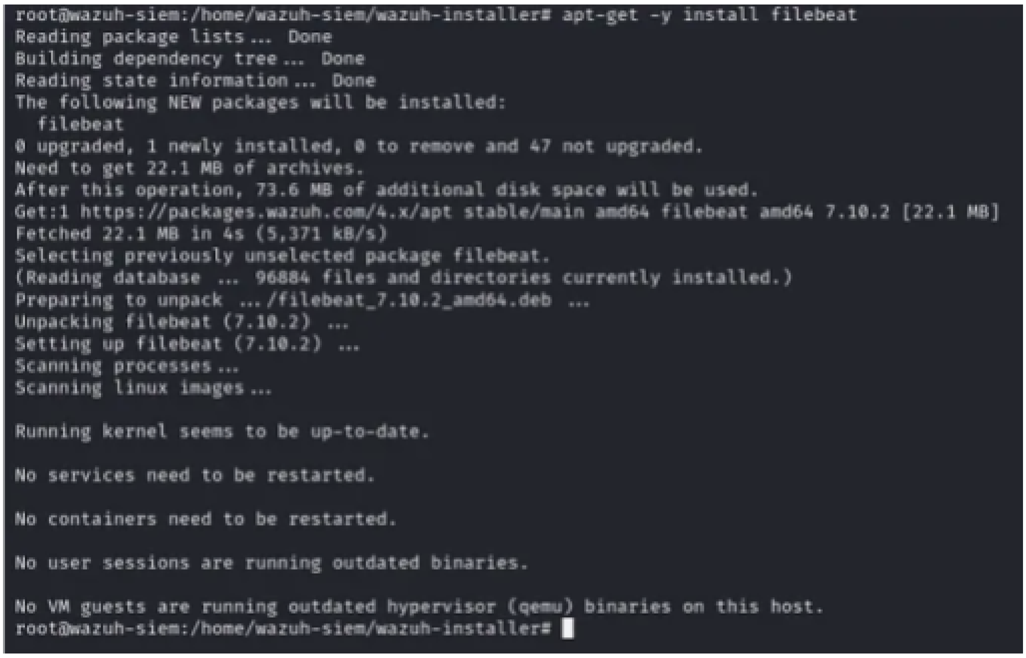
```
network.host: "192.168.251.150"
node.name: "node-1"
cluster.initial_master_nodes:
- "node-1"
#- "node-2"
#- "node-3"
cluster.name: "wazuh-cluster"
#discovery.seed_hosts:
#  - "node-1-ip"
#  - "node-2-ip"
#  - "node-3-ip"
node.max_local_storage_nodes: "3"
path.data: /var/lib/wazuh-indexer
path.logs: /var/log/wazuh-indexer

plugins.security.ssl.http.pemcert_filepath: /etc/wazuh-indexer/certs/indexer.pem
plugins.security.ssl.http.pemkey_filepath: /etc/wazuh-indexer/certs/indexer-key.pem
plugins.security.ssl.http.pemtrustedcas_filepath: /etc/wazuh-indexer/certs/root-ca.pem
plugins.security.ssl.transport.pemcert_filepath: /etc/wazuh-indexer/certs/indexer.pem
plugins.security.ssl.transport.pemkey_filepath: /etc/wazuh-indexer/certs/indexer-key.pem
plugins.security.ssl.transport.pemtrustedcas_filepath: /etc/wazuh-indexer/certs/root-ca.pem
plugins.security.ssl.http.enabled: true
plugins.security.ssl.transport.enforce_hostname_verification: false
plugins.security.ssl.transport.resolve_hostname: false

plugins.security.authcz.admin_dn:
- "CN=admin,OU=Wazuh,O=Wazuh,L=California,C=US"
plugins.security.check_snapshot_restore_write_privileges: true
plugins.security.enable_snapshot_restore_privilege: true
plugins.security.nodes_dn:
- "CN=node-1,OU=Wazuh,O=Wazuh,L=California,C=US"
#- "CN=node-2,OU=Wazuh,O=Wazuh,L=California,C=US"
#- "CN=node-3,OU=Wazuh,O=Wazuh,L=California,C=US"
plugins.security.restapi.roles_enabled:
- "all_access"
- "security_rest_api_access"

plugins.security.system_indices.enabled: true
plugins.security.system_indices.indices: [".plugins-ml-model", ".plugins-ml-task", ".opendistro-alerting-config", ".opendistro-alerting-alert*", ".opendistro-anomaly-results*", ".opendistro-anomal

### Option to allow Filebeat-oss 7.10.2 to work ###
compatibility.override_main_response_version: true
```

Figure 8: Running Status



```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# systemctl status wazuh-indexer
● wazuh-indexer.service - Wazuh-indexer
     Loaded: loaded (/lib/systemd/system/wazuh-indexer.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2024-01-14 16:45:00 UTC; 2min 1s ago
       Docs: https://documentation.wazuh.com
   Main PID: 117618 (java)
      Tasks: 58 (limit: 4515)
     Memory: 1.3G
        CPU: 54.310s
     CGroup: /system.slice/wazuh-indexer.service
             └─117618 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss1m -Djav
```

Figure 9:Installing Filebeat.



```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# apt-get -y install filebeat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  filebeat
0 upgraded, 1 newly installed, 0 to remove and 47 not upgraded.
Need to get 22.1 MB of archives.
After this operation, 73.6 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 filebeat amd64 7.10.2 [22.1 MB]
Fetched 22.1 MB in 4s (5,371 kB/s)
Selecting previously unselected package filebeat.
(Reading database ... 96884 files and directories currently installed.)
Preparing to unpack .../filebeat_7.10.2_amd64.deb ...
Unpacking filebeat (7.10.2) ...
Setting up filebeat (7.10.2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#
```

Figure 10: Configuring Filebeat

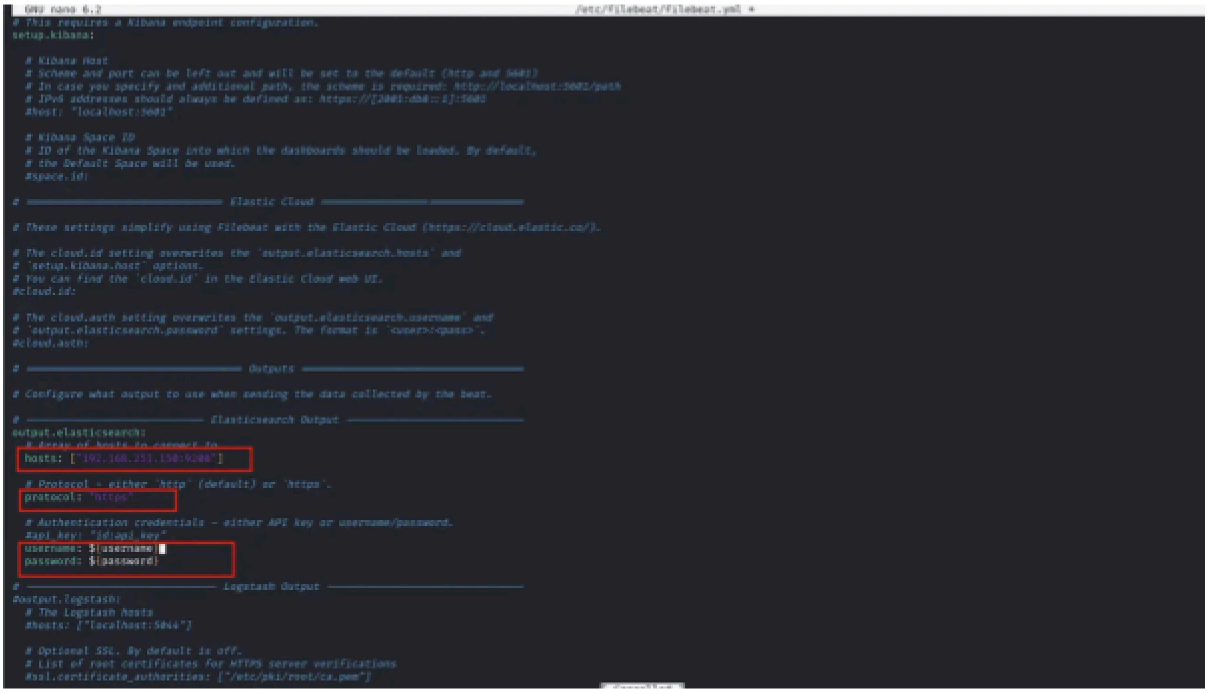


Figure 11: Filebeat running status

Figure 12: Wazuh Dashboard

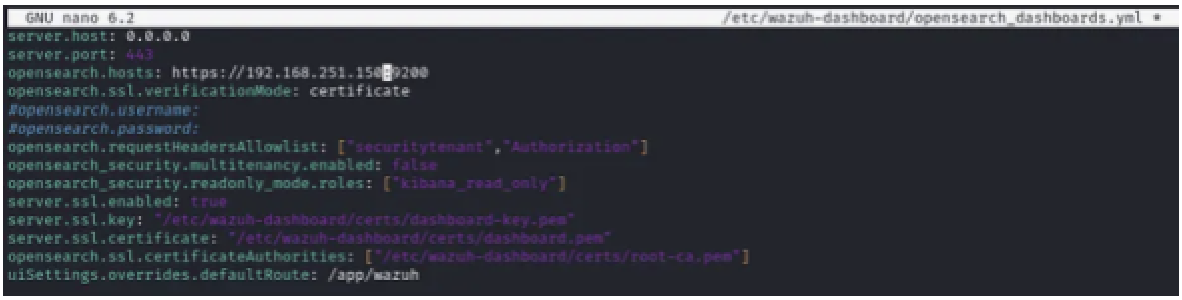
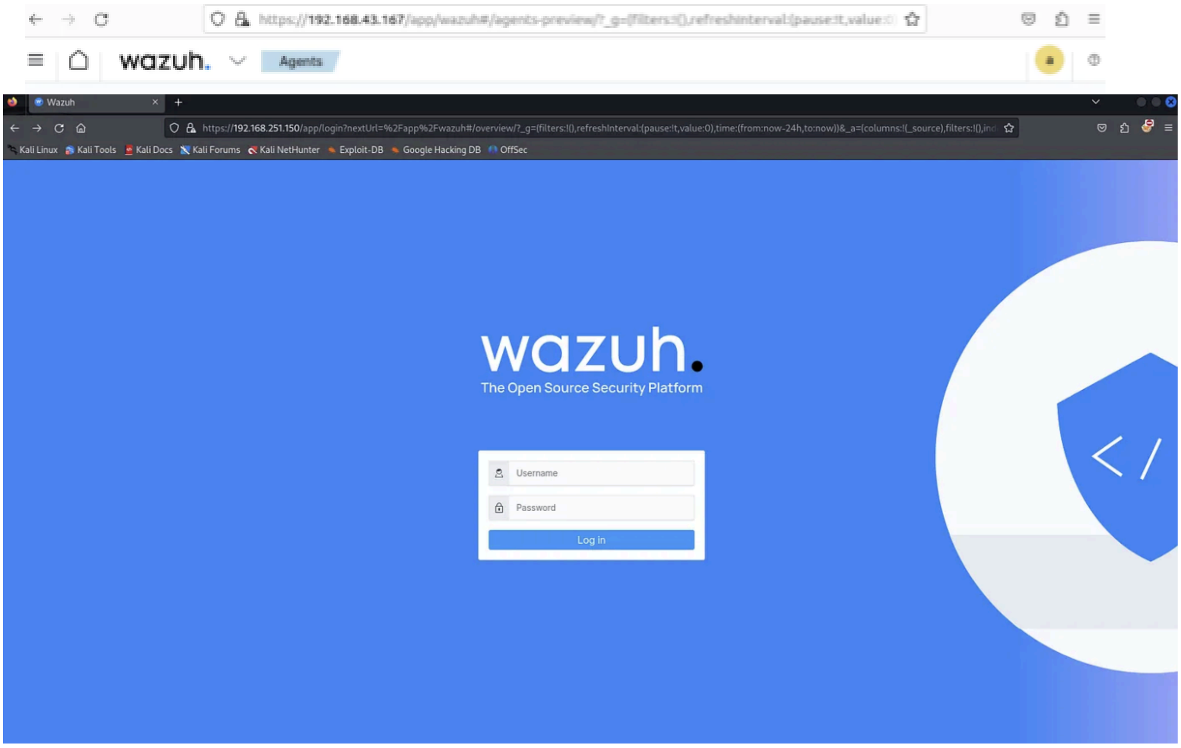


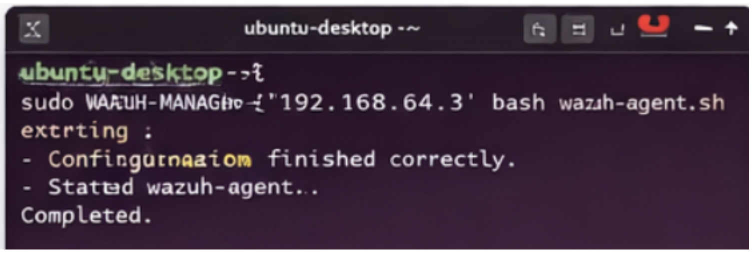
Figure 13: Final Output



Week 3&4: Threat Detection and Security Monitoring.

- Wazuh – EDR / SIEM platform
- OpenSearch – log indexing and analytics
- Kibana – visualization dashboards
- Hydra – brute force simulation tool
- Atomic Red Team – adversary attack simulation framework

The goal of Week 3&4 is to verify that the SOC environment can **detect suspicious activities and security threats** using the Wazuh monitoring platform.



At this stage:

• The Wazuh Manager is installed
• The Wazuh Agent is connected to the endpoint
• Logs are being collected from the endpoint system

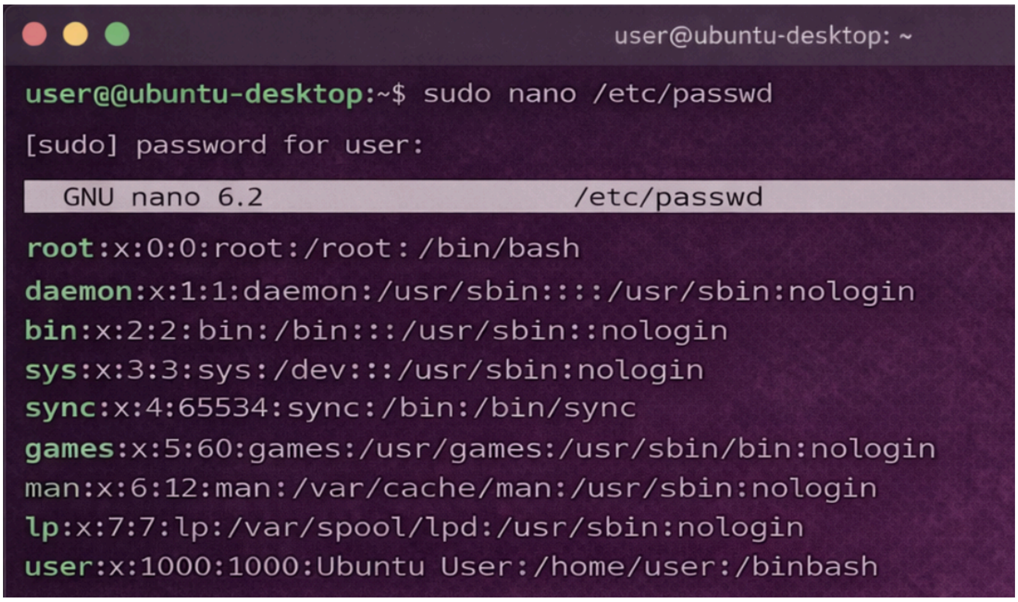
Now the SOC system begins **monitoring and analyzing security events in real time**.

2. File Integrity Monitoring (FIM)

One of the most important features of Wazuh is **File Integrity Monitoring (FIM)**. FIM monitors critical system files and detects any unauthorized changes.

Examples of monitored files:

• `/etc/passwd`
• `/etc/shadow`



• `/etc/sudoers`

3. Log Monitoring:

This log records:

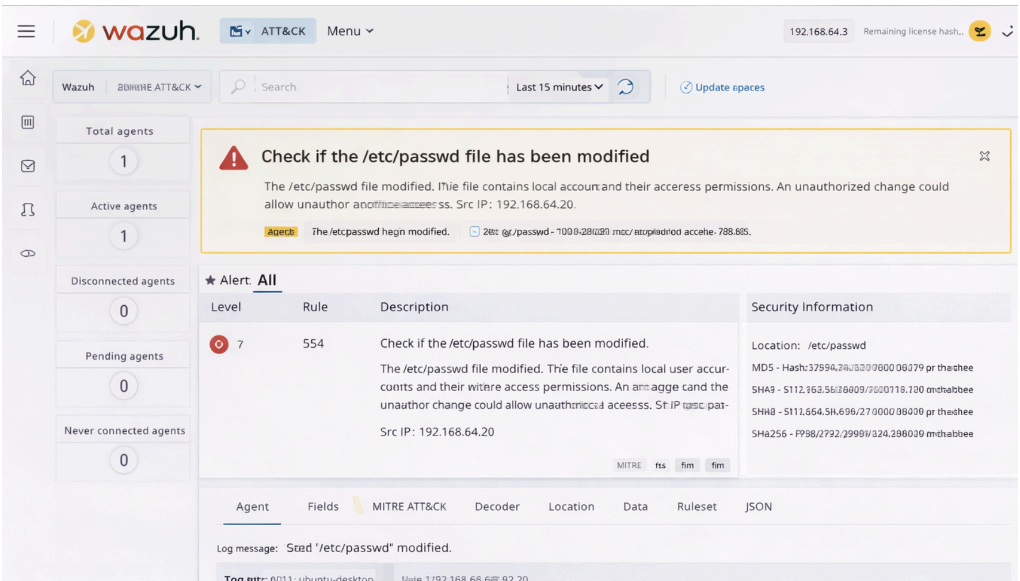
• Login attempts
• Failed authentication
• Privilege escalation attempts

Monitoring these logs helps detect attacks such as:

• brute force login attempts
• unauthorized access
• suspicious user activity

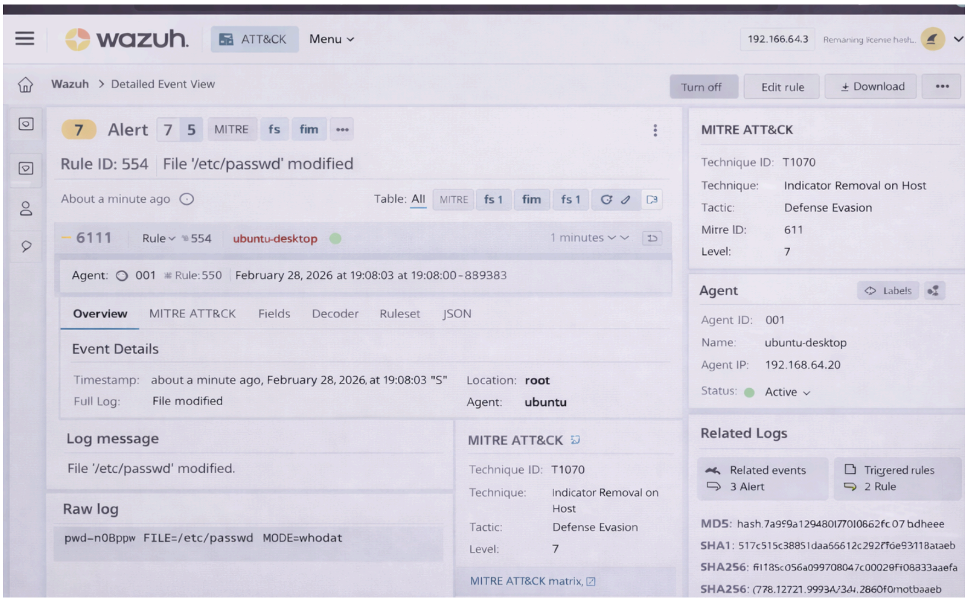
hydra -l user -P rockyou.txt ssh://192.168.64.3 (attack using this command)

• `hydra` → password cracking tool
• `-l` → username
• `-P` → password list
• `ssh://` → target service

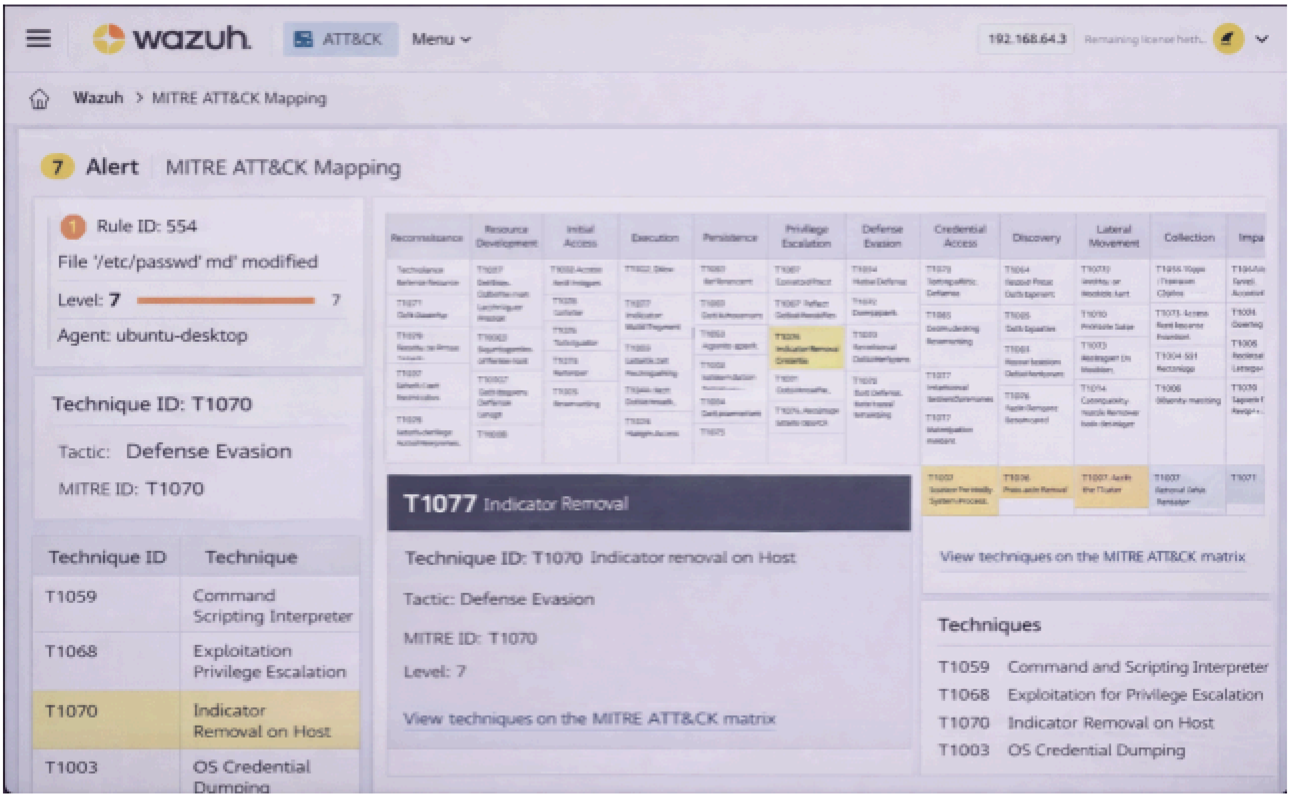


4. Incident Investigation

- Event logs
- Source IP address
- Affected endpoint
- User activity



clear visual example of a MITRE ATT&CK mapping screen



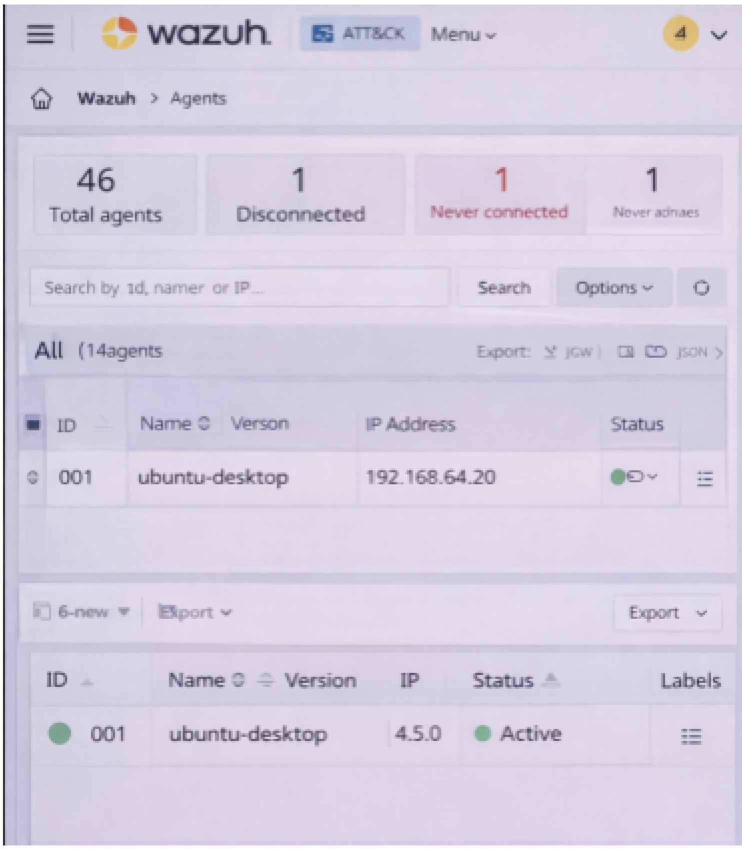
MITRE ATT&CK Mapping

- Maps alerts to attacker techniques
- shows attack stages and tactics
- helps analysts understand attacker behavior.

Agents Dashboard

- displays all monitored endpoints
- shows agent status and IP address
- enables real-time endpoint security monitoring.

Dashboard Overview



The project successfully demonstrated:

1. Detection of SSH brute force attacks.
2. Automatic blocking of attacker IP.
3. Simulation of ransomware techniques.
4. Visualization of attack lifecycle in SOC dashboard.

Screenshots collected:

- Hydra brute force attack
- Wazuh alert logs
- Firewall blocking rule
- Kibana attack visualization

Advantages

Sentient Shield provides:

- Real-time threat detection
- automated incident response
- centralized log monitoring
- threat hunting capabilities
- MITRE ATT&CK mapping

Conclusion

This project demonstrates the development of an enterprise SOC monitoring system capable of detecting and responding to cyber threats in real time. By integrating **Wazuh**, **OpenSearch**, and **Kibana**, the system provides automated security monitoring and response.

The implementation of active response and threat simulation highlights the effectiveness of the **Sentient Shield EDR Grid** in identifying and mitigating modern cyber attacks.